# Assignment No.1

Q.1 Explain Booth algorithm for signed binary multiplication

## Booth Algorithm

It is an efficient method used to perform multiplication of signed binary numbers (positive and negative numbers). It reduces the number of addition/substraction operations and handles 2's complement representation easily.

## Idea

Instead of adding multiplicant repeatedly Booth's algorithm analyses bit pairs of the multiplier

## Registers used

A - Accumulator
Q - Multiplier
M - Multiplicand
Q-1 - Extra bit
Count - Number of bits

## Rules

| Q0 | Q-1 | operation |
|---|---|---|
| 0 | 0 | No operation |
| 0 | 1 | Add M to A |
| 1 | 0 | Substract M from A |
| 1 | 1 | No operation |

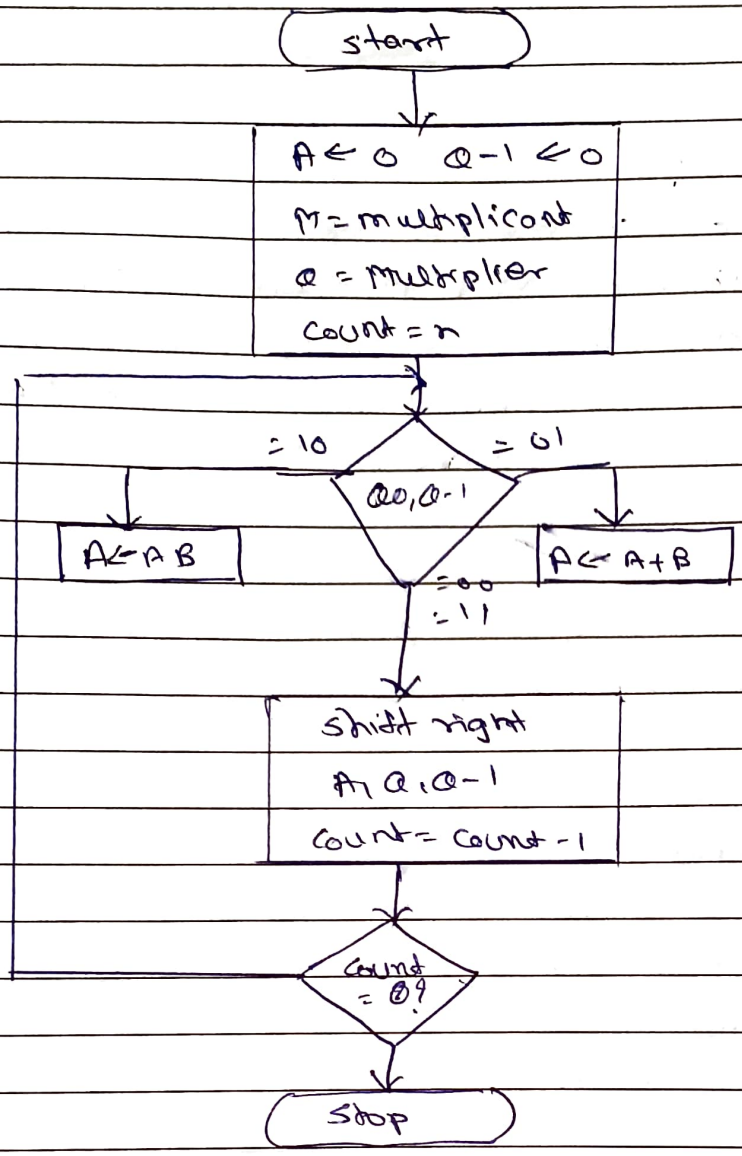


fig. Booth's Algorithm

Q2. Explain the working of carry look ahead adder and analyze how it improve speed compared to a ripple carry adder

Carry look ahead adder improves addition speed by reducing delay caused by carry propagation in ripple adder,

It is a fast binary adder used in digital circuits to reduce the delay caused by carry propagation during addition. Instead of waiting for each carry to move step by step from one bit to the next (as in ripple carry adder), CLA calculates carries in advance using logic equations

CLA uses two signals

① Generate - Indicate whether a bit pair will produce carry
$G = A \times B$

② Propagate - Indicates whether incoming carry will pass forward.
$P = A + B$

Formular to calculate carry
$C1 = G0 + P0\,C0$
$C2 = G1 + P1\,C1 = G1 +$
$C3 = G2 + P2\,C2 =$
$C4 = G3 + P3\,C3 =$

Advantages
- Much faster than ripple adder
- Reduces propagation delay
- High speed arithmetic circuits
- Used in CPUs and ALU design

Disadvantages
- More hardware required

Q.3 Compare von Neumann and Harvard architectures

| Aspects | Von Neumann | Harvard |
|---|---|---|
| Memory | single memory for data and instruction fields | There is seperate memory for data and instructions |
| Bus | It has the single bus | It has seperate buses |
| Speed | Its speed is slower | Its speed is faster |
| Complexity | simple | complex |
| cost | ~~Higlow~~ | High |
| parallel access | No | Yes |

Q.4 Apply the 2's complement method to compute the addition and substraction of following signed binary numbers

Addition

a) +28 and −18

+25 + (-18)

25 = 11001

18 = 10010 → 2's complement = 01110

= 11001
+ 01110
1 00111

carry ignore

Answer = 00111 = +7

b] -14 + (-9)

14 = 01110 → 2's complement = 10010

9 = 01001 → 2's complement = 10111

= 10010
+ 10111
1 00001

ignore carry

= 01001 = -23

Substraction

a) +25 - (-18)

25 = 11001

18 = 10010 → 2's complement = 01110

= 11001
ignore carry 10010
1 01011

$= 101011 = 43$

b] $-14 - (-9)$

$14 = 01110 \rightarrow$ 2's complement $= 10010$

$9 = 01001 \rightarrow$ 2's complement $= 11011$

$$= \begin{array}{r} 10010 \\ + \; 11011 \\ \hline 11011 \end{array}$$

$= 11011 = -5$

Q.5. Describe the interconnection structure of a computer system and explain the role of buses.

Interconnection structure refers to communication pathways connecting CPU, memory, and I/O devices

Main Components

1. CPU
2. Memory
3. Input / Output device
4. system Bus

Types of Buses

1. Data Bus
- Transfers actual data
- Bidirectional

- example: data between RAM and CPU

② Address Bus
- carries memory address
- Unidirectional (CPU - Memory)

③ Control Bus
- Carries control signals
- Example read/write signals

Functions
- Communication between components
- Data Transfer
- Control Coordination
- Address Identification

Advantages
- Efficient communication
- organized data flow